

UMC 0.11um RFCMOS Al Advanced Enhancement SPICE MODEL DOCUMENT (Bonding Pad RF Model)

**Version: 0.2 Phase 1
2010/7/22**

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1 Introduction

1.1 Revision History

Table 1-1 Revision History

Version	Date	Owner	Comment
V0.2p1	07/22/2010	Ji Wei Hsu	1. Shrinking from L130 bonding pad RF model 2. Base on new DSM definition From 2009/11/30

1.2 General Information

This document describes Bonding Pad RF models for UMC's 110nm process.

The compact model is provided in various formats. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

The high-frequency responses for the RF Bonding Pad models in this document are generated based on microwave scattering-parameter (S-parameter) measurements. The S-parameter measurement system is well-calibrated through a ceramic impedance standard substrate (ISS) to ensure measurement accuracy.

The models should be used in conjunction with UMC capacitor Pcells (Programmable layout Cell) provided in UMC MM/RF FDK (Foundry Design Kit). The models may not be valid if the capacitor layout is different from UMC Pcell.

In current release, 3 major commercial simulators (Cadence's SPECTRE, Synopsys' HSPICE, and Mentor Graphics' ELDO) are supported.

It is suggested to read the model document first to understand the model usage and limitation before using the models.

The devices couldn't be put under Bond Pad for UMC's 110nm process.

Some minimum size (W and L) couldn't be chose, therefore the size of Bond Pad chose refer TLR

1.3 Bonding Pad RF Sub-circuit Model

Model File Name:

Spectre:

L110AE_pad_RF_V021.lib.scs

L110AE_pad_RF_V021.mdl.scs

HSPICE:

L110AE_pad_RF_V021.lib

L110AE_pad_RF_V021.mdl

Eldo:

L110AE_pad_RF_V021.lib.eldo

L110AE_pad_RF_V021.mdl.eldo

Sub-circuit Model Name & Nodes:

Pad_rf (top sub) -- Pad over psub

Corner Section:

FF: minimum capacitance value

TT: typical capacitance value

SS: maximum capacitance value

Input Parameters:

index_layer: selection index for metal layer (default is 8)

index_layer=8: 1P8M process è with Al, M8 + sub (ground)

index_thick: selection index for metal thickness (default is 1)

index_thick=1: 40K

index_pad: selection index for Pad type (default is 1)

index_pad=1: 50um BOAC padè pad width =53um, pad length=80.2um

1.4 Model Usage

Synopsys HSPICE release 2007.09
Cadence SPECTRE version 6.2.1.299
Mentor Graphics ELDO version 6.11_1.1

1. For Cadence Spectre

Copy model files into your simulation directory

Add the following statement in the input netlist to the simulator:

include “./L110AE_pad_RF_V021.lib.scs” section=CornerSection

The CornerSection could be typ, min or max

Follow the following example to call the mdoel:

x1 (top sub) pad_rf index_layer=8 index_thick=1 index_pad=1

2. For Synopsis Hspice

Copy model files into your simulation directory.

Add the following statement in the input netlist to the simulator

.lib “./L110AE_pad_RF_V021.lib ” CornerSection

, where CornerSection could be typ, min, max.

Follow the example below to call the model

x1 top sub pad_rf index_layer=8 index_thick=1 index_pad=1

3. For Mentor Graphics Eldo

Copy model files into your simulation directory.

Add the following statement in the input netlist to the simulator

.lib “./L110AE_pad_RF_V021.lib.eldo ” CornerSection

, where CornerSection could be typ, min, max.

Follow the example below to call the model

x1 top sub pad_rf index_layer=8 index_thick=1 index_pad=1

2 Bonding Pad

2.1 Scope:

In this report, the RF modeling results of Bonding Pad are described.

2.2 Device Architecture and Layout:

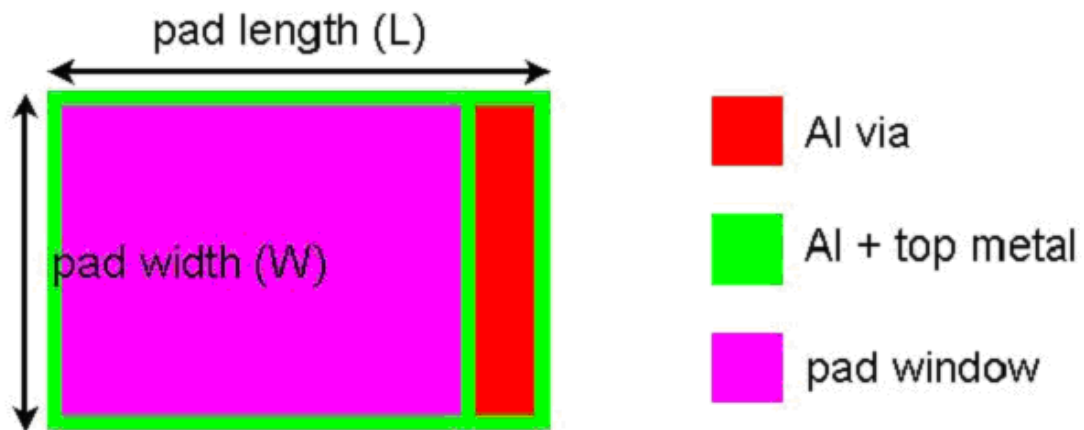


Figure 2-1 Illustration of Bonding Pad over psub

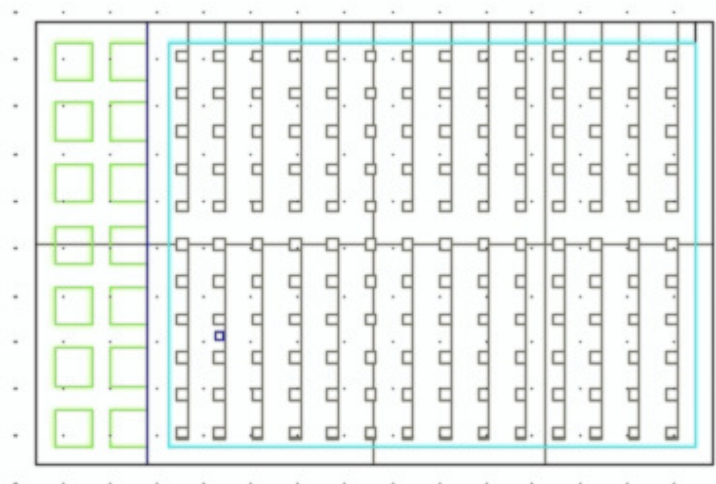


Figure 2-2 Example layout for Bonding Pad

2.3 Equivalent Circuit:

The equivalent circuit of Pad sub-circuit model is shown in Fig 2-1. The capacitance between top metal and psub is represented by C_{ox} . R_{si} and C_{si} represent the equivalent resistance and capacitance of the psub under bond Pad, respectively.

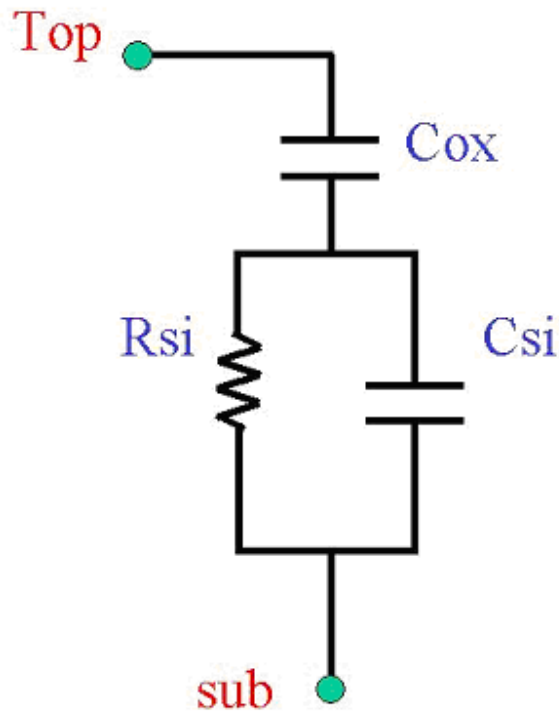


Figure 2-3 sub-circuit for RF Bonding Pad model over psub

2.4 Model Scaling Range (before 90% shrinkage)

Pad Width (W): 53 μ m
Pad Length (L): 80.2 μ m
Frequency range: 0 ~ 20GHz

2.5 Comparison between model & measurement over frequency

Effective shunt capacitance: $C_{ox} = -1/\{\omega \cdot \text{Im}[1/Y_{11}]\}$, ω is angular frequency

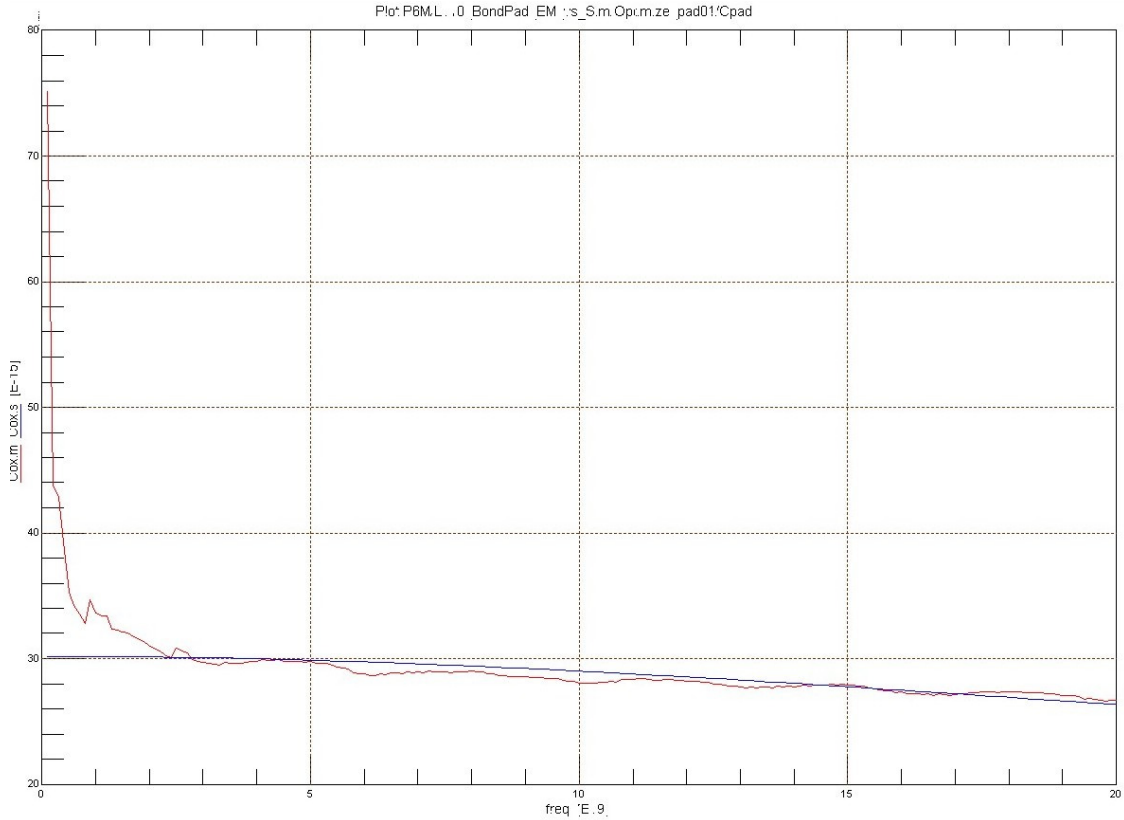


Figure 2-4 Measurement vs Model

Figure 2.4 Frequency response for 1P8M, 40K, w=53um l=80.2um Bonding Pad

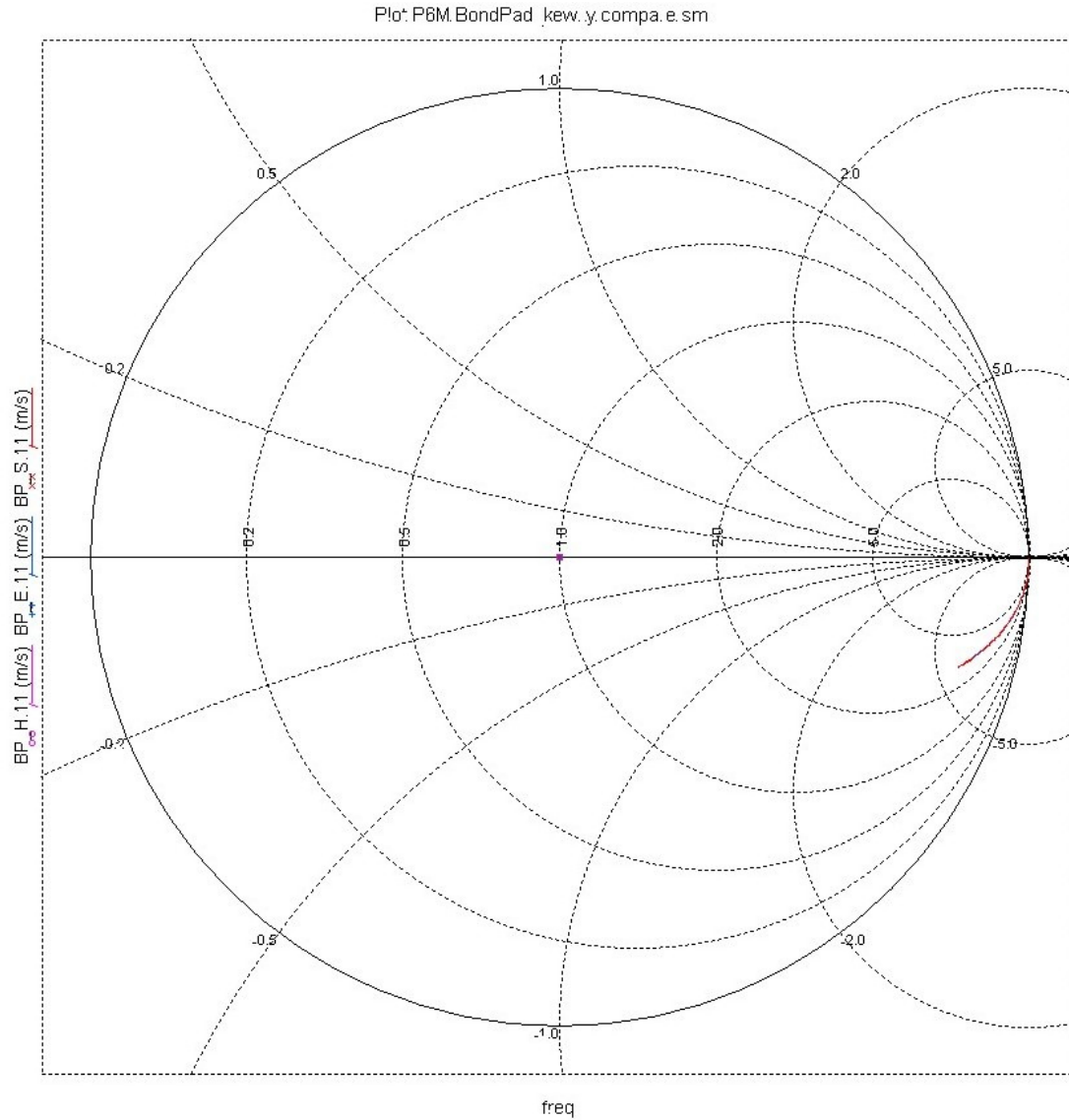


Figure 2-5 Smith Chart for Spectre Hspice Eldo

Figure 2.5 Frequency response for 1P8M, 40K, w=53um l=80.2um Bonding Pad